Origins & Evolution: The Master Study Notes

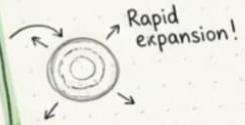
A comprehensive guide from the Big Bang to Homo Sapiens

(Exam prep! Focus on Miller's experiment & Hardy-Weinberg!)

Module 1 : The Origin of the Universe & Earth

Step 1: The Big Bang Theory

A singular, unimaginable huge explosion. The universe expanded and cooled. Hydrogen (H) and Helium (He) formed. Gases condensed under gravitation to form galaxies.



Step 2: Earth Forms (4.5 Billion Years Ago)

Initially a molten mass. There was NO atmosphere. Water vapour, methane (CH_4), carbon dioxide (CO_2), and ammonia (NH_3) were released from the surface.

Step 3: Atmospheric Chemistry

UV rays from the sun broke water into Hydrogen and Oxygen. The lighter H_2 escaped. Oxygen combined with ammonia and methane to form water and CO_2.

Step 4: Cooling & Oceans

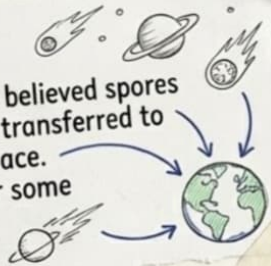
The ozone layer formed. As Earth cooled, water vapour fell as rain to fill fill depressions, creating oceans.

Life finally appeared 500 million years after the Earth's formation (almost 4 billion years back).

Early Theories of Life

Panspermia

Early Greek thinkers believed spores or units of life were transferred to Earth from outer space. (Still a favourite for some astronomers!).



Spontaneous Generation

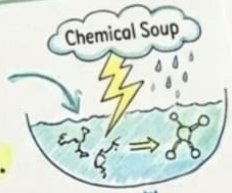
The belief that life came out of decaying and rotting matter like straw and mud.

Louis Pasteur's Refutation

Demonstrated that life comes only from pre-existing life. His experiment showed that in pre-sterilised flasks, life did not come from killed yeast, whereas in open flasks, living organisms arose. Spontaneous generation was dismissed!

Oparin & Haldane – The Winners

Russia & England.
Proposed that the first life came from pre-existing non-living organic molecules (RNA, proteins).



Chemical evolution preceded biological evolution!

Biological Evolution

Module 2: The Shift in Evolutionary Thought

Theory of Special Creation

3 Connotations (all challenged in the 19th century):

- ☑ 1. All living organisms were created as such.
- ☑ 2. Diversity was always the same and will remain so.
- ☑ 3. Earth is only about 4,000 years old.

Darwin & Wallace's Breakthrough

Charles Darwin: Travelled on H.M.S. Beagle. Observed that existing life forms share similarities not only among themselves but with extinct life forms.

Alfred Wallace: Worked in the Malay Archipelago and came to similar conclusions simultaneously.

Darwinian Fitness. This refers ultimately to reproductive fitness. Those better fit in an environment leave more progeny, survive more, and are selected by nature (Natural Selection).

Drivers of Evolution: Lamarck vs. Malthus

Lamarckism (French Naturalist)	Thomas Malthus (Populations)
• **Mechanism:** Evolution driven by the use and disuse of organs. • **Classic Example:** Giraffes elongated their necks in an attempt to forage tall trees. • **Outcome:** Inheritance of acquired characters (passing this elongated neck to succeeding generations). Conjecture no longer believed!	• **Mechanism:** Natural resources are limited. Population size will grow exponentially if everyone reproduces maximally, but in reality, population sizes remain stable. • **Outcome:** Competition for resources. Only some survive and flourish. Darwin based natural selection on this fact! Variations that enable better resource utilisation mean only those adapted reproduce and leave progeny.

The Missing Link: Mutations & Saltation

The Problem: Darwin observed variations but kept silent on the inheritable 'factors' influencing phenotype.

Hugo de Vries: Based on his work on the evening primrose, he brought forth the idea of mutations.

Mutations – Large differences arising suddenly in a population.

He believed it is mutation which causes evolution & speciation, not minor variations (heritable) that Darwin talked about.

Mutations are random and directionless while Darwinian variations are small and directional.

Darwin vs. de Vries

Darwinian Variations:	de Vries Mutations:
Small, directional, and gradual.	Large differences arising suddenly, random, and directionless.

Saltation – A single step, large mutation that causes speciation.

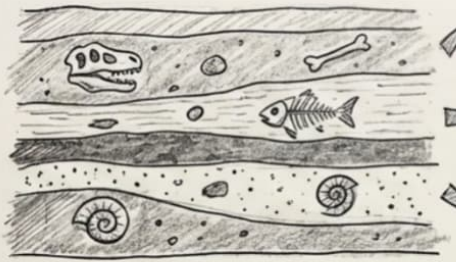
random, directionless

evolution was gradual

Sometimes single step large mutations called saltation that causes evolution.

Module 3: Physical Evidences for Evolution

Palaeontological Evidence



- Fossils are remains of hard parts of life-forms found in rocks.
- Rocks form sediments; different-aged rock sediments contain different life-forms that died during formation.
- Shows new forms of life have arisen at different geological time periods (e.g., Dinosaurs).

Embryological Evidence

- **Ernst Haeckel**: Proposed support based on embryonic features absent in adults (e.g., all vertebrate embryos develop a row of vestigial gill slits behind the head).
- **Karl Ernst von Baer's Objection**: Careful study showed embryos never pass through the adult stages of other animals.

Homology vs. Analogy Matrix

	Concept	Origin & Evolution	Classic Examples
Homologous Organs	Same anatomical structure, different functions.	Common ancestry (Divergent Evolution).	Mammal forelimbs. Whales, bats, cheetahs, and humans share similarities in the pattern of bones (humerus, radius, ulna, carpals, metacarpals, phalanges).
Analogous Organs	Different anatomical structures, same function.	Different ancestors adapting to similar environments (Convergent Evolution).	Wings of butterfly vs. bat; flippers of penguins vs. dolphins; sweet potato (root) vs. potato (stem).

organic evolution

manic anatomi structures

Module 4: Natural Selection in Action

Case Study: Industrial Melanism (Moths in England)

Before Industrialisation

Tree trunks covered in white lichen. White moths survived by blending in; dark moths were spotted and eaten.

After Industrialisation

Tree trunks became dark with soot. White moths stood out and were eaten. Dark moths survived via predator evasion.

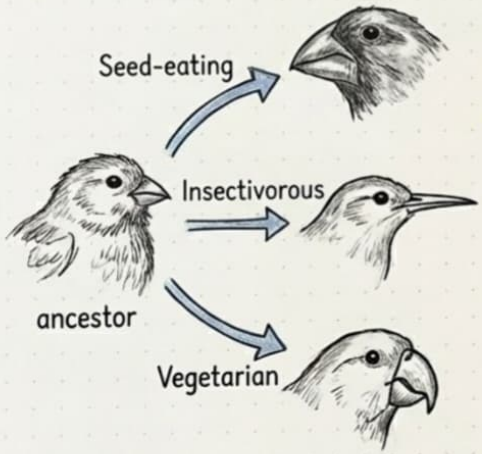
Conclusion: Nature selects for fitness based on the prevailing conditions.

Evolution can also happen via anthropogenic action! E.g., Use of herbicides and pesticides has resulted in the evolution of resistant varieties of insects in a fraction of human time scales.

Geography & Evolution: Adaptive Radiation

Adaptive Radiation: The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography.

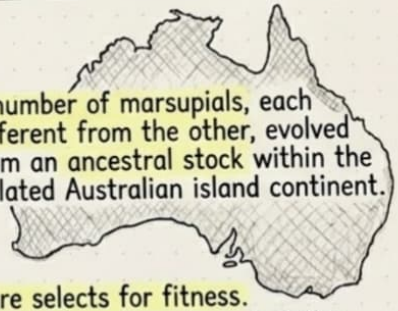
Case 1: Darwin's Finches (Galapagos Islands)



From an original seed-eating ancestor, many different beak forms evolved.

Adaptation allowed them to become insectivorous and vegetarian finches based on available food.

Case 2: Australian Marsupials



A number of marsupials, each different from the other, evolved from an ancestral stock within the isolated Australian island continent.

Nature selects for fitness. Fitness is based on characteristics which are inherited. Adaptive ability is inherited and has a genetic basis. Fitness is the end result of the ability to adapt and get selected by nature.

* (Convergent evolution note: When multiple adaptive radiations occur in an isolated geography, placental mammals evolve to look exactly like their corresponding marsupial rat/anteater!)

Module 5: Population Genetics & Hardy-Weinberg

The Principle: Allele frequencies in a population are stable and constant from generation to generation (Genetic Equilibrium).

Frequency of homozygous dominant individuals (AA)

Frequency of homozygous recessive individuals (aa)

$$\boxed{p^2 + 2pq + q^2 = 1}$$

Frequency of heterozygous individuals (Aa)

The sum total of all allelic frequencies.

When the frequency measured differs from expected values, the difference indicates the extent of evolutionary change.

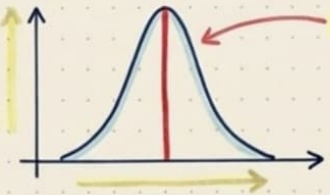
Disrupting the Genetic Equilibrium

The 5 Factors Affecting Hardy-Weinberg:

1. Gene flow / gene migration
2. Genetic drift
3. Mutation
4. Genetic recombination
5. Natural selection

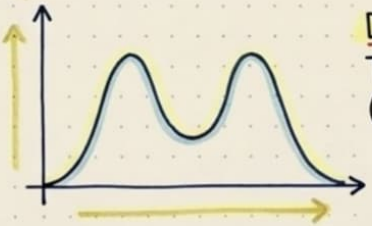
The Founder Effect

When a small drifted population becomes entirely different, the original drifted population becomes founders of a new species.



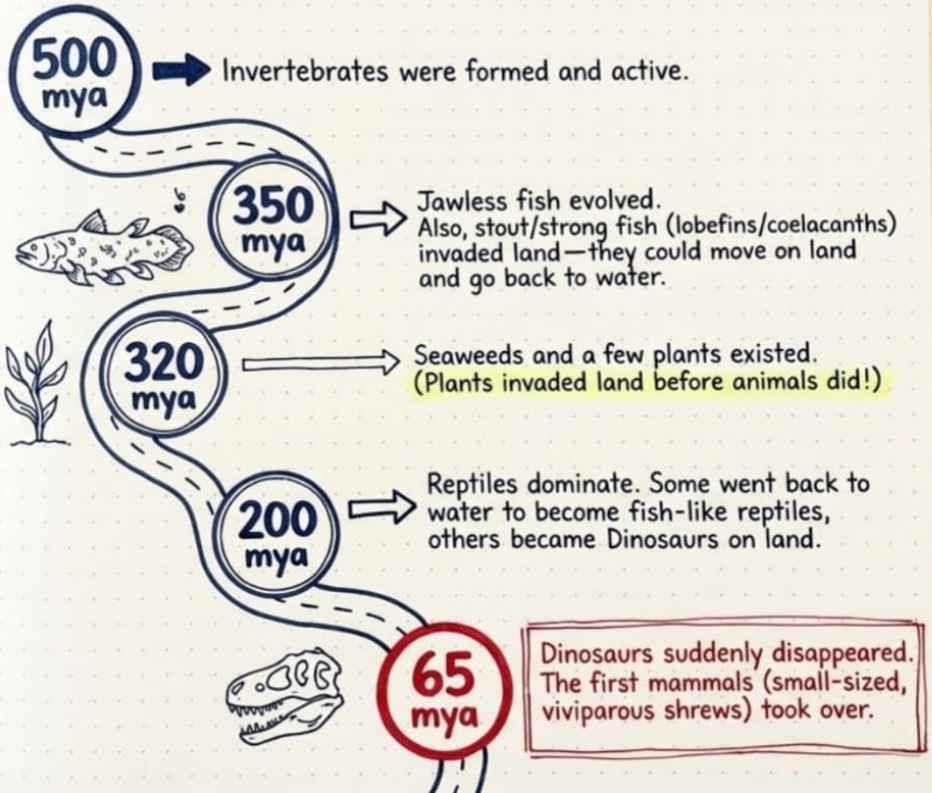
Stabilising: Peak gets higher and narrower (favours intermediate traits).

Directional: Peak shifts in one direction (favours one extreme).



Disruptive: Two peaks form (favours both extremes over the intermediate).

Module 6: The Geological Timeline



The Ancestry of Man (Part 1)

15 mya → **Dryopithecus & Ramapithecus**

- Traits: Hairy, walked like gorillas/chimpanzees.
- Distinction: Dryopithecus was more ape-like, while Ramapithecus was more man-like.

3-4 mya → **Australopithecines**

- Location: East African grasslands.
- Traits: Hunted with stone weapons but essentially ate fruit.

2 mya → **Homo habilis (The First Human-like Being)**

- Brain Capacity: 650 - 800 cc.
- Traits: Did not eat meat. First hominid.

The Ancestry of Man (Part 2)

1.5 mya

Homo erectus

- Brain Capacity: 900 cc.
- Traits: Probably ate meat.

100k - 40k years ago

Neanderthal man

- Brain Capacity: 1400 cc.
- Location: Lived in near east and central Asia.
- Culture: Used hides to protect their body and buried their dead.

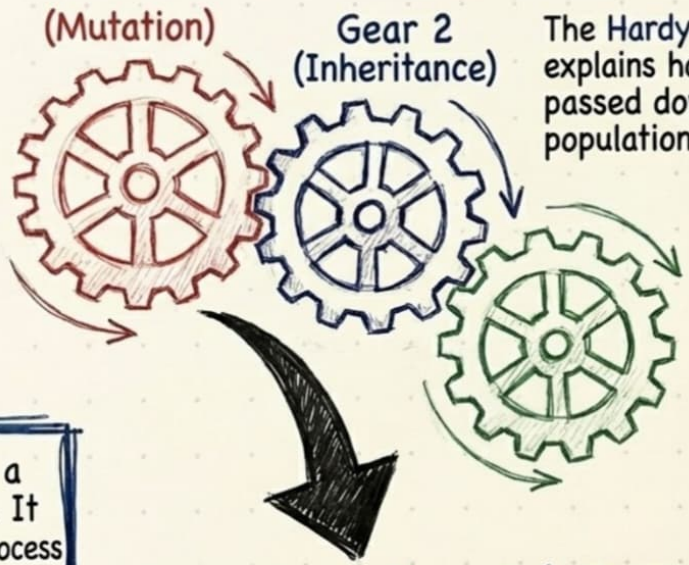
75k - 10k years ago

Homo sapiens

- Origin: Arose in Africa and moved across continents into distinct races.
- Culture: Pre-historic cave art developed about 18,000 years ago. Agriculture and settlements started around 10,000 years back.

The Grand Unification: How it all fits together

Gear 1 (Mutation)
Hugo de Vries' Saltation provides the sudden, raw genetic variation.



The Hardy-Weinberg Principle explains how these alleles are passed down (or drifted) through populations over generations.

Gear 3 (Environment)
Adaptive Radiation and isolated environments (like the Moths or Finches) apply survival pressure.

> Evolution is not a directed process. It is a stochastic process based on chance events in nature and chance mutations in organisms!

The Output (Darwin's Brilliance):

Genetic Variation + Inheritance + Environmental Pressure

= **Natural Selection & Branching Descent.**